

Mira Askar

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EDUCATION

University of Warwick, Coventry, United Kingdom

BEng Electrical and Electronic Engineering | Expected graduation: July 2028 | Grade(first year): First-Class Honours

WORK EXPERIENCE

Healware | London, United Kingdom (Remote) | Java Backend Developer | Aug – Sep 2026

- Developed Java/Spring RESTful API workflows and endpoint contracts to ingest medication selections and patient allergy histories from hospital systems via structured JSON payloads.
- Designed response DTOs and API schemas to return allergy-risk assessments, contributing factors and safer medication alternatives to downstream clinical systems.
- Defined input-validation rules, exception-handling flows, HTTP status codes and structured error responses to support reliable exchange of sensitive clinical data.

Trans Asia Construction LLP | Kyzylorda, Kazakhstan | Java Backend Developer | Jul – Aug 2026

- Developed a Java/Spring employee-record backend using Hibernate ORM and PostgreSQL, containerised with Docker, to centralise staff profiles and employment data.
- Designed relational data models and CRUD workflows for employee records and document-status tracking, enabling structured search and administration.
- Implemented server-side validation, database queries and update workflows to improve data consistency and reduce manual file-based record management.

PROJECTS

Hyperloop University Project | Electronics Hardware Engineer | Sep 2025–Present

- Simulated and validated circuit behaviour in LTspice before schematic capture and PCB layout, preparing the design for physical prototyping.
- Designed a custom battery-temperature monitoring PCB in KiCad, including the schematic and board layout
- Incorporated CAN and UART interfaces for communication between the temperature-monitoring system and the Battery Management System.

Poker Decision Simulator | Java

- Developed a Texas Hold'em simulator with complete betting rounds, hand evaluation and AI-controlled opponents.
- Implemented Monte Carlo simulation to estimate hand equity and support decisions under incomplete information.
- Built an explainable fold/call/raise engine using pot odds, expected value, object-oriented design and JUnit tests.

Delivery Robot & Navigation Platform | Java, Spring Boot, PostgreSQL, ESP32, GPS | In Progress

- Developing an ESP32-based robot with GPS positioning, motor control, obstacle detection and wireless telemetry.
- Building a Spring Boot REST API for robot control, route planning, delivery management and live status monitoring.
- Designing a PostgreSQL data model for robots, deliveries, GPS waypoints, telemetry and route history, with ESP32-backend integration.

Interactive Pixel-Art Portfolio Website | Frontend Development

- Designed and developed a responsive pixel-art portfolio with a retro desktop UI, custom sprite animations and cross-browser compatibility for Safari and Firefox.

ACHIEVEMENTS

- British Physics Olympiad (BPhO) - Gold; Gold CREST Award for QinetiQ project.
- University of Warwick Global Excellence Scholarship; Malvern College Honorary Scholarship.
- Secured approximately £1,000 in sponsorship from KT Lab Cloud and Chevron at the Enactus National Expo Kazakhstan.